RB18 Stats

- Wins: 17
- Podiums: 28
- Poles: 8
- Points: 759

Christian Horner
Team Principal, Red Bull Racing

Max Verstappen
F1 Driver, Red Bull Racing

Sergio Perez
F1 Driver, Red Bull Racing

Adrian Newey
CTO, Red Bull Racing

Picture: Red Bull. Composite: Kamesh Raj Shanmugam

RED BULL'S 2022 DOMINANCE: KEY DESIGN DECISIONS

Story by: Matt Somerfield

The year 2022 marked the long-awaited debut of the extensive changes originally planned for the previous year, which were delayed due to the COVID-19 pandemic. The changes included tighter financial regulations, which constrained spending, and revised sporting regulations that reshaped race weekends. But perhaps the most significant changes were in the technical regulations, as F1 introduced a new generation of cars designed to promote closer racing and more excitement.

Despite not starting the year as the fastest car, the RB18 had to overtake Ferrari's F1-75. The car was piloted by defending world champion Max Verstappen and Sergio Pérez. With Verstappen behind the wheel, the RB18 secured his second consecutive Drivers' Championship, while Red Bull claimed their fifth Constructors' Championship, their first since 2013.

Adrian Newey's design for the RB18 is his most successful F1 design to date and is considered one of the most dominant Formula One vehicles ever constructed. However, the RB18's success is not just due to some magic bullet, but a combination of factors that work together to create a superb overall package.

Suspension

Red Bull has been a trailblazer in the world of Formula 1 suspension systems from 2009 to 2021. Their ingenious implementation of a pull-rod rear suspension and a push-rod front suspension in the RB5 car revolutionized the sport. This suspension layout was found to be the most aerodynamically superior under the then-existing regulations. Eventually, other teams followed suit and adopted Red Bull's novel approach. However, for the 2022 season, Red Bull has made some notable changes to their suspension system for their RB18 car.

They have opted for a push-rod layout at the rear to accommodate the larger diffuser, and a pull-rod setup at the front. The front suspension retains a single-element wishbone design, but with the setup flipped over from before. As part of the team's 2022 repackaging efforts, the steering assembly, previously buried in the chassis, has been relocated to its natural position at the front of the bulkhead. Red Bull's pursuit of innovation and optimization has led them to once again make bold changes to their suspension system.

"While there were some exceptions, such as the testing of the pull-rod method at the front end, Red Bull's suspension layout remained the standard for over a decade"

Picture: Red Bull

Sebastian Vettel fears F1 retirement could leave him in a deep 'hole'

Vettel is set to enter 2023 with uncertainty about what lies ahead.The 35-year-old has announced he is to retire from F1 racing at the end of this season, having made his debut in the 2007 United States Grand Prix just a few weeks before his 20th birthday. After eight races for Sauber in that first season, the following year Vettel joined Toro Rosso and in only the 22nd F1 race of his career he captured his first victory at the 2008 Italian Grand Prix. The German went on to join Red Bull in 2009, and from 2010-13 won the World Championship four consecutive times, eventually taking his number of race wins to 53 when joining Ferrari. But when he completes his second and final year with Aston Martin, Vettel will step into the unknown one way or another – even if he does land himself another job in the remaining months of this year. It is feasible that Vettel may instead focus on the environmental and social issues about which he frequently speaks out.

Picture: Red Bull

Bib Tray

When it comes to designing a Formula 1 car, one of the most crucial aspects is the arrangement of sprung components that provide compliance with the car's suspension. Red Bull has taken a unique approach in this area by employing a Belleville spring in the bib region of their RB18 car. This carefully chosen element offers several distinct advantages over traditional suspension setups. One key advantage of the Belleville spring is its compact size, allowing for a more efficient installation within the car's design. This feature is particularly important in Formula 1, where every millimetre of space counts towards improving performance. The Belleville spring also has the added benefit of reducing the amount of vibration that the car experiences, leading to smoother handling and improved overall performance on the track. In addition to the Belleville spring, the RB18 also features several supporting beams and internal stays that help to maintain the floor's rigidity relative to the chassis. This added support reduces the amount of flex on the floor as the load is imparted upon it, leading to improved handling and faster lap times. Red Bull's dedication to innovation and optimization is evident in their careful selection of sprung elements for their Formula 1 cars. The choice of a Belleville spring for the bib region, along with the inclusion of supporting beams and internal stays, are just a few examples of the team's attention to detail and commitment to pushing the boundaries of technological advancement in the sport.

"It is worth noting that this choice of suspension setup may be one of the factors that contribute to Red Bull's superior performance in terms of reduced porpoising and bouncing when compared to its rivals"

The Floor

With the new regulations implemented in Formula 1, the importance of the floor and diffuser's performance has become even more critical, with changes in ride height affecting various aspects of their design. However, the added complexity of these components has also led to a significant increase in weight. In an attempt to limit the extreme solutions witnessed on the floor's edge in previous years, the FIA introduced new rules in 2021. Red Bull, in particular, decided not to employ the allowable 'edge wing' as some of its competitors did but instead opted to derive inspiration from the regulations themselves. Consequently, a horseshoe-shaped cutout was introduced ahead of the Z-shaped cutouts, resulting in a split level in the floor. The shorter section located behind is positioned at a different height and has a distinct contour compared to the section ahead. Red Bull also modified the upper surface of the floor and sidepod by adjusting the size and shape of the blister and channel located on the edge of the side pod. The design of the underfloor on the RB18 is particularly interesting since Red Bull did not settle for conventional shapes and forms. For instance, the floor's bow section features unconventional contouring that works hand-in-hand with the forward strake, making it a unique design compared to many other teams. This innovative design has since been adopted by Red Bull's rivals. The 'ice skate' solution, which was initially introduced on the RB18, has also been incorporated by other teams. Red Bull's commitment to staying ahead of the competition is evident in their continuous pursuit of unique designs and optimization of their car's critical components like the floor and diffuser.

"Red Bull has made multiple efforts to optimize the portion of the floor, adding a Gurney-like flap to the corner of the floor's front section for improved performance"

Brake Cooling

Red Bull Racing has made iterative developments on the inner front brake duct fairing in the opening phase of the current season. This is due to the increase in the size of wheel rims, from 13 inches to 18 inches, which has resulted in a larger diameter for the brake discs by about 50mm. Nevertheless, the drill holes that make their way from the centre of the disc out to their edge are more limited, allowing a minimum diameter of only 3mm. As a result, cooling holes are no longer permitted in the pads. This change has a significant impact on the design of numerous components, including the airflow and heat exchange of the assembly. To address this issue, Red Bull and several of its rivals have developed an internal brake disc fairing that isolates the brake disc inside a much larger drum, thereby changing the passage of air into, around, and away from the brake disc to help moderate the temperatures that will ensue.

To maintain optimal brake temperature, Red Bull Racing has made frequent changes to the fairing design and its contents. The original design of the insulation packing surrounding the inner face of the disc has been replaced by a carbon fibre panel. Additionally, the colour of both the fairing and the brake calipers has changed to feature a thermal coating that aids in the transfer of heat. These iterative changes demonstrate the Red Bull team's attention to detail and their commitment to improving their car's performance throughout the season.

The development of the internal brake disc fairing is just one of the many design decisions that have allowed Red Bull to dominate the 2022 season. The team has had to overcome various challenges, including the increase in the size of wheel rims and the new regulations that have put even more emphasis on the performance of the floor and diffuser.

Body Work

Red Bull Racing has been working tirelessly throughout the season to update the RB18's sidepod and engine cover bodywork. The aim is to achieve a better balance between the car's cooling requirements and its aerodynamic output, which is essential for success in Formula One racing. One of the most significant updates to the RB18 was introduced at the British Grand Prix. This update involved the implementation of a shelf-like shoulder section that extends rearwards from the halo and cockpit transition. This new design creates an upper ledge that the airflow naturally follows and also serves as a divisional line for the competing flow structures below. Following the British Grand Prix, the team continued to refine the design of the RB18. At the following race in Austria, they removed the rearmost section of the shark fin to open up the rear section of the engine cover.

Another refinement was made for the Belgian Grand Prix. This time, the team focused on adjusting the mid-section bodywork between the engine cover shelf and the sidepod drop off. As a result of these changes, the airflow's approach into the coke bottle region just aft of these revisions was met with a change to the shape of the lower wishbone fairing. The lower wishbone fairing now features an indent in the upper surface of the floor to accommodate its travel.

This latest update ensures that the airflow is able to approach the coke bottle region in an optimal manner, further enhancing the car's performance. The team's commitment to optimizing the car's aerodynamic output has resulted in a series of significant updates, culminating in the latest refinement to the lower wishbone fairing. These updates demonstrate the team's dedication to continually improving the RB18's performance to remain competitive throughout the season.

Picture: Red Bull

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Wing Solutions

The FIA's rule changes for 2022 have significantly reduced the advantages teams can gain from the design of their front and rear wings. However, there is still a lot of variation up and down the grid, with the ratio between downforce generation, balancing the car, and flow conditioning still critical in delivering the required performance for each team.

The biggest change was made at the Australian Grand Prix, where the diveplane on the outside of the endplate was replaced with an S-shaped variant to produce a different pressure gradient response. Red Bull has been particularly proactive with its beam wing arrangement. Starting the season with a stacked bi-plane arrangement, it has since trimmed the elements to suit its overall downforce and drag targets. The beam wing connects the flow structures at the rear of the car, influencing the diffuser and rear wing. Red Bull has used polar-opposite configurations for circuits requiring the most and least downforce. For lower downforce venues, Redbull has removed the upper of the two beam wing elements. The RB18 is one of the most efficient cars in the field, as shown by its speed trap numbers compared to the level of wing it is running when compared to its rivals. Consequently, the team has not focused on a low drag biased rear wing design, even for Monza. In fact, it abandoned attempts to trim the upper flap's trailing edge to reduce drag when it induced a similar DRS oscillation issue that the team had faced towards the end of its 2021 campaign.

> *"To optimize performance, the RB18's front wing upper flap has been adjusted several times to trim the car front-to-rear for specific circuit characteristics"*

Picture: Vladimir Rys

Diverging Interests

Not every development that is added to an F1 car results in improvement. Teams often have to weigh the positives and negatives of pursuing a design that doesn't immediately offer the expected yield. It's also true that what's seen as a solution to provide the ultimate lap time for the car doesn't work for both drivers. Red Bull introduced a development at the British Grand Prix that is causing some headaches. Max Verstappen's RB18 has been fitted with a plain solution, while Sergio Perez has run a more complex arrangement - similar to what Ferrari has used for some time now. Due to the cost cap and a reduction in the use of CFD and wind tunnel time based on championship position, teams have to be more careful than ever in managing the development of this year's car and next year's challenger. As a result, we're likely to see fewer developments arrive during the end of the season. However, some of the research conducted into next year's car could yield a meaningful uplift in performance that warrants full-scale parts being manufactured■.

Pirelli have announced their intention to continue on as the official tyre supplier to Formula 1 for the 2025 season onwards, with the sport having recently put out a tender inviting applicants for the deal.

The Italian manufacturer took over from Bridgestone in supplying Formula 1's tyres from 2011, when they were initially brought in on a remit of creating tyres which would degrade more quickly than their predecessors, prompting more pit-stops and a variety of strategies as a result. The working relationship between Pirelli and Formula 1 has now been established for more than a decade, with the tyre manufacturer most recently overseeing the sport's change onto 18-inch wheels and the introduction of a sixth dry compound for the 2023 season. But with their contract up at the end of next season, Formula 1 has invited applications for tyre suppliers for the 2025 season onwards, which would encompass the Formula 2 and Formula 3 championships as well. Now that the technical aspects of the tyres have been outlined, the sport's current supplier have confirmed their interest in continuing their current arrangement – with an overall decision over a bid to come in due course. In a statement published on Wednesday morning, Pirelli said: "The FIA has now published the invitation to participate in the next tender process for the supply of tyres to the top single-seater championships – FIA Formula One World Championship, Formula 2 and Formula 3 – for the three year period from 2025-2027 (with an option for 2028). "The document outlines technical characteristics that are broadly in line with the tyres used today and their relevance to technology transfer from track to road, putting a particular emphasis on sustainability. "The framework described by the FIA is closely aligned to Pirelli's motorsport strategy and so is of great interest, with the Italian company having been Global Tyre Partner to the sport for more than a decade, since 2011.

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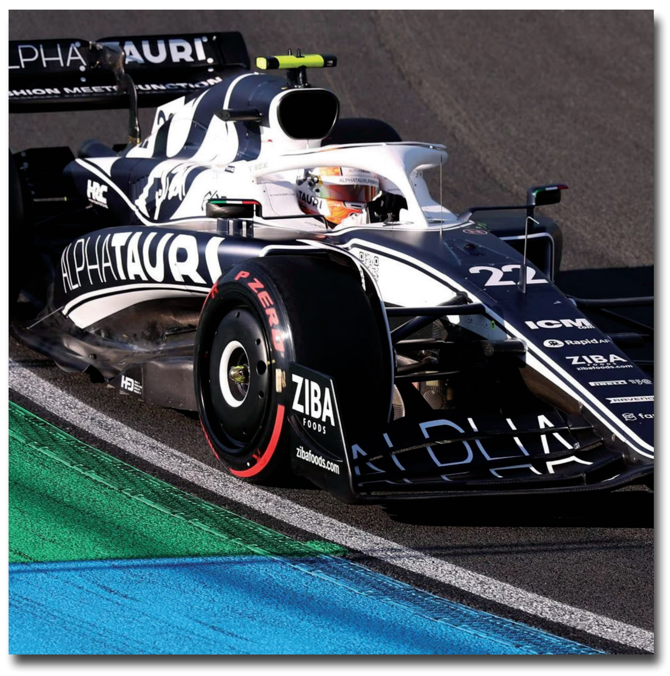

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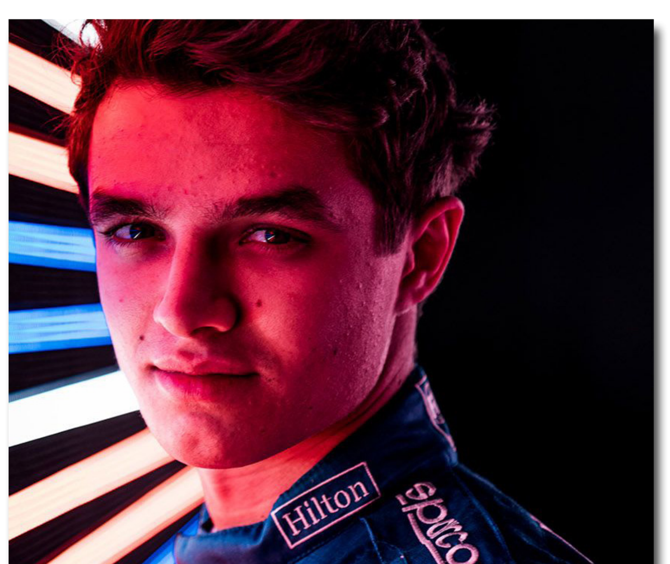

THE KEY F1 DESIGN DECISIONS THAT ALLOWED RED BULL TO DOMINATE 2022

By: Matt Somerfield, Co-author: Giorgio Piola

March 23, 2023

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The year 2022 marked the long-awaited debut of the extensive changes originally planned for the previous year, which were delayed due to the COVID-19 pandemic. The changes included tighter financial regulations, which constrained spending, and revised sporting regulations that reshaped race weekends. But perhaps the most significant changes were in the technical regulations, as F1 introduced a new generation of cars designed to promote closer racing and more excitement. Despite not starting the year as the fastest car, the RB18 had to overtake Ferrari's F1-75. The car was piloted by world champion Max Verstappen and Sergio Pérez. With Verstappen behind the wheel, the RB18 secured his second consecutive Drivers' Championship, while Red Bull claimed their fifth Constructors' Championship, their first since 2013. However, the RB18's success is not just due to some magic bullet, but a combination of factors that work together to create a overall package.

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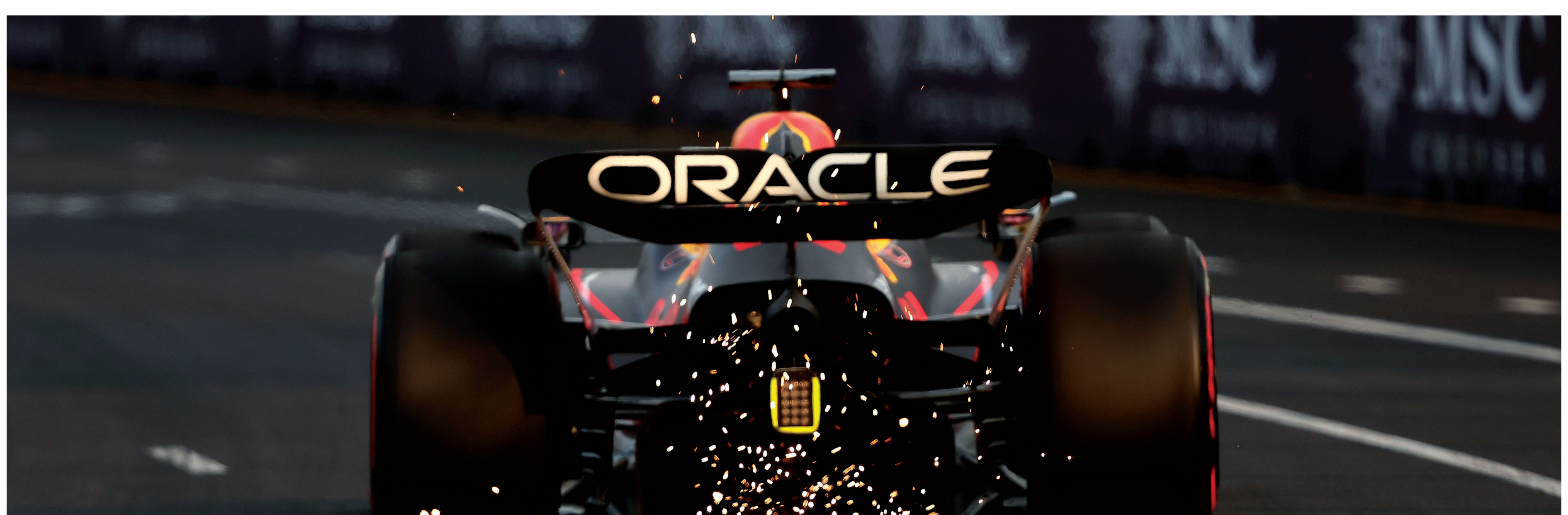

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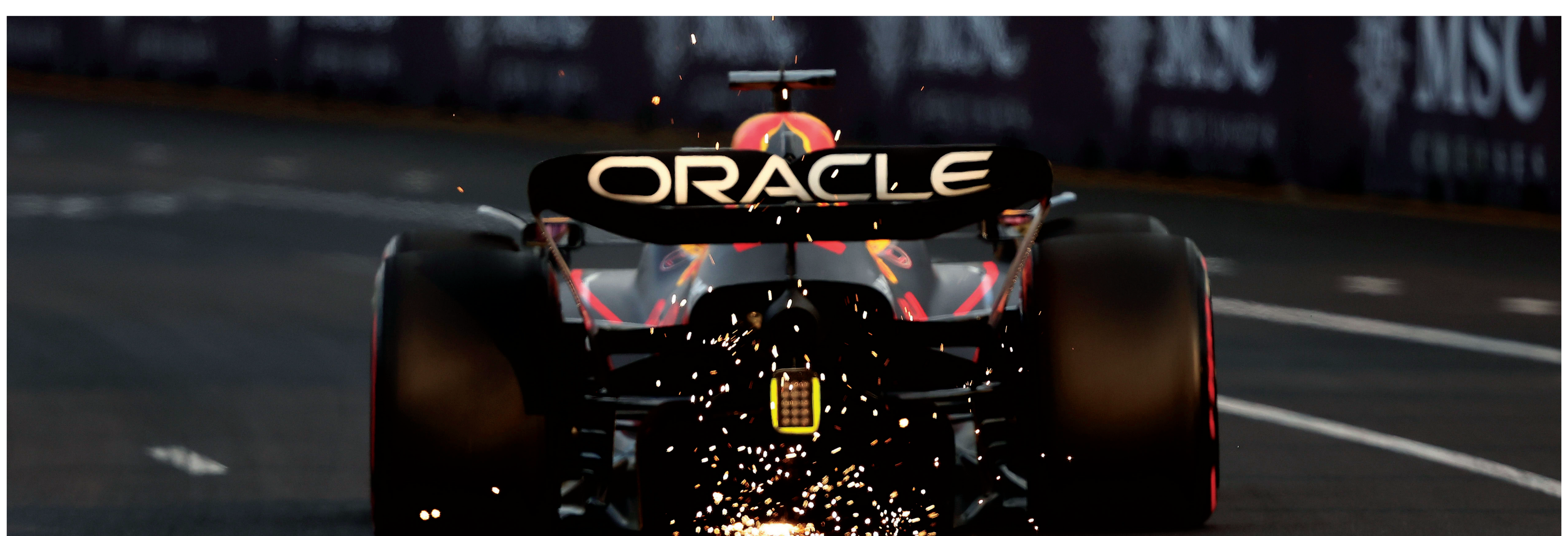

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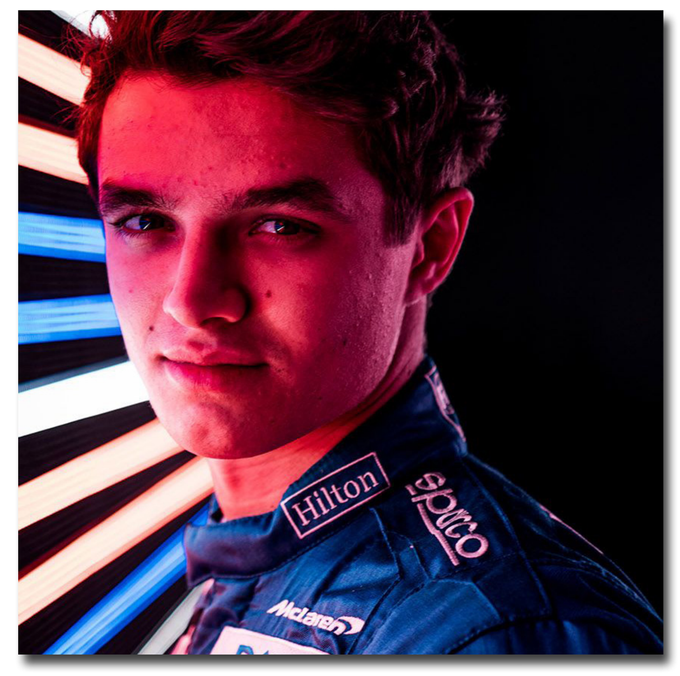

Lando Norris Snubs Daniel Ricciardo In Bold Statement

The goal of the feature is to make the reader engage with the content. The tone of the story is technical and informative. The layout of the magazine inside is conventional. The choice of headlines, pull quotes and 50% text - 50% high quality imagery is used to keep the audience engaged throughout the story. There are side stories in the feature to give the readers a break from the original story yet keeping them focused on reading the magazine. The color scheme in the magazine reflects the team colors'.

The online adaption of the feature is simple. Clean and minimalist layout with little to no color on the website. It makes it easier for the reader to navigate without any distractions.

Link to the article,
https://www.motorsport.com/f1/news/red-bull-key-f1-design-choices-dominate-2022/10375323/